
When Are Recursive Language Models Useful? A Cost-Aware, Task-Conditional Evaluation

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Abstract

1 Recursive Language Models (RLMs) handle long inputs through controller-
2 generated programs and additional model calls. Published RLM evaluations
3 typically compare these scaffolds with generic long-context or coding baselines.
4 Deployment requires a different comparison: whether adaptivity improves on
5 routes suited to the task under a common information, scoring and budget contract.
6 We evaluate the released standard RLM against deterministic operators, retrieval,
7 semantic workers and direct inference in an operation-matched audit. On controlled
8 operations and known-family Oolong aggregation, matched routes generally match
9 or outperform standard RLM at lower cost. A source-transfer replication preserves
10 this direction although its clustered exact-rate difference includes zero. Open
11 multi-document search is less stable. An initial untouched-shard replication favors
12 RLM over BM25 but neither a prospectively locked equal-cap comparison nor a
13 post-hoc same-row repeat yields an RLM contrast that survives multiplicity correc-
14 tion. Route ordering changes across these rollouts while RLM remains less reliable
15 and $5.8\text{--}15.9\times$ costlier than the stronger alternatives. Mechanism probes localize
16 errors to operation specificity, output-schema fit, evidence selection and scaffold
17 reliability. These results support reporting recursive inference as a task-conditional
18 accuracy–cost frontier rather than as a single scaffold ranking.

19 1 Introduction

20 Long-context systems can retrieve, compress, apply fixed operators, delegate local judgments,
21 use direct inference or generate adaptive programs. Strong benchmark performance by adaptive
22 programming establishes capability but not deployment value. A deployment decision also depends
23 on the information and tools available to each route, cost, completion and ranking stability.

24 RLMs [Zhang et al., 2025] make this distinction consequential. A controller writes Python over
25 externalized context and may recursively query models. The original paper compares this general
26 route mainly with task-agnostic long-context and coding scaffolds and reports strong OOLONG-
27 Pairs results [Zhang et al., 2025, Bertsch et al., 2026]; follow-ups add self-reflective search, typed
28 combinators and recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Lumer et al., 2026].
29 Deployed alternatives include RAG–long-context routing [Li et al., 2024, Chen et al., 2026a], retrieval
30 [Lewis et al., 2020], prompt compression [Jiang et al., 2023], deterministic operators and decomposed
31 workers [Khot et al., 2023]. The relevant comparison is RLM against the strongest affordable route
32 permitted by the task contract.

33 We evaluate this deployment decision directly. For a stated task family, information contract and
34 budget, we ask whether adaptive program generation yields a stable accuracy–cost advantage over
35 operation-matched alternatives. This extends cost-aware agent auditing [Kapoor et al., 2025, Yehudai
36 et al., 2026, Chen et al., 2025b, Bragg et al., 2026] and differs from RLM trajectory diagnosis: the

Table 1: Controlled task-family construction. Each family targets a different long-context operation with a prespecified alternative route.

Family	Operation	N	Expected cheap mechanism
T-STRUCT	Exact pair enumeration	30	SLM extraction + fixed operator
T-MIXED	Directional semantic counting	30	Direct LLM / local judgments
T-NLP (diagnostic)	Standard classification	24	SLM direct

unit is the deployed route, not recursion depth or one scaffold component. Declared specialization, paired uncertainty and rollout stability determine the credibility of the resulting task-conditional route frontier [Yu and Kumbier, 2020].

We compare released standard RLM with deterministic, retrieval, semantic-worker and direct routes. Oolong and an RLM trajectory already decompose labeling from aggregation [Zhang et al., 2025, App. E.2]; our study adds a same-row comparison of complete deployed routes. Table 3 summarizes the boundary. We contribute:

1. An operation-matched protocol that fixes rows and scorers, declares route-specific prior knowledge and reports accuracy, failures, latency, calls and cost. An answer-contract repair separates finalization failure from route quality.
2. Wrong-family, same-operator, finalization and trace controls that distinguish operation knowledge from tool access and localize differences in semantic judgment, output schema and scaffold reliability; prior RLM trajectory diagnoses do not combine these controls [Zhang et al., 2025, Wang, 2026].
3. A task-conditional deployment boundary: known-operation routes lead on Oolong under paired tests, while BrowseComp route ordering changes across same-row rollouts and no RLM contrast survives Holm correction. The resulting two-tier recommendation shows why a single scaffold score is insufficient for deployment.

2 Operation-Matched Evaluation

Scope. The audited system is the official `rlms v0.1.3` standard inference-time RLM: a controller generates Python over externalized context and may call additional models before answering. We audit this public, installable implementation against operation-matched alternatives on the same instances. Direct standard-RLM accuracy rows cover T-STRUCT, T-MIXED and the Oolong slices; T-NLP is a compatibility diagnostic. Claims remain scoped to this implementation; tuned depth, reasoning or stronger-family variants may move the frontier. We also run MIT OASYS’s official post-trained RLM-Qwen3-30B-A3B policy and the unmodified public λ -RLM code with its Qwen3-8B tier on frozen CD-15 under the same input/scorer contract. These are same-contract diagnostics, not reproductions of the official SRLM, λ -RLM or RAH benchmark protocols; adapter-based rows remain pressure tests only.

Tasks. Table 1 defines three controlled operation families. Public evidence adds LongBench-v2 (LB2) Academic QA ($N=94$) [Bai et al., 2025], label-provided Oolong-synth ablations ($N=80, 150$), standard-unlabeled Oolong slices ($N=30, 120, 150, 45$) and BrowseComp-Plus estimates on an original frozen $N=80$ manifest, a stopped attempted- $N=49$ replication and two fixed- $N=45$ untouched-query endpoints [Chen et al., 2025a]. The second 45-row endpoint prospectively adds iterative lexical search and equal decision/time caps across all four routes. SU-150 spans 8k–131k tokens but overlaps previously evaluated windows; CD-45 is balanced over 1k, 4k and 262k tokens and excludes prior IDs/windows, while VC-45 uses the official validation split with two upstream corpora absent from all previous Oolong runs. Oolong exposes no source-document ID so neither 45-row slice supports document-disjointness claims.

Table 2: Model roles and their corresponding evaluation settings.

Role	Model	Use
RLM ctrl.	Gemini 2.5 Flash; GPT-5; Qwen3.6-35B-A3B	Standard scaffold; direct matched Gemini; BrowseComp matched routes
Trained/code RLM	Qwen3-30B-A3B; λ /Qwen3-8B	Frozen CD-15 diagnostics
Direct LLM	Gemini 2.5 Flash; Qwen3.6-35B-A3B	Direct-context and matched-controller baselines
SLM worker	Gemma 4 26B-A4B	Direct, CoT, labeling and fallback routes
Cheap controller	GPT-4o-mini	Deliberately weak cheap controller; T-NLP orchestration only

Evaluation design. The controlled families cover exact aggregation, semantic counting and classification; label-provided Oolong separately probes output-schema inference. We then evaluate public slices chosen to probe settings in which RLM should be competitive. This design trades leaderboard breadth for trace-level analysis: headline rows are tied to frozen inputs, cost logs, contract labels and leakage/split checks. Figure 1 declares what each route is allowed to know before any comparison is run. The non-RLM routes intentionally use task-family prior knowledge, fixed operators and prompt/schema engineering; we treat this as deployment specialization to account for, not as a scaffold-only causal intervention. Our recommendation would be overturned by an official trained policy or same-contract SRLM/ λ -RLM/RAH reversal, a wrong-family typed route that still wins or an RLM route with the same operator library matching cost and accuracy.

Baselines and model roles. Table 2 lists the role each model plays. Baselines include BM25, iterative lexical search, textual decomposition, fixed operators, Gemma 4 26B-A4B workers, LLM+SLM routes, direct Gemini 2.5 Flash and standard RLM. The primary Oolong RLM controller is Gemini 2.5 Flash; T-STRUCT uses GPT-5 and the BrowseComp-Plus matched-controller audit uses Qwen3.6-35B-A3B for all routes. Appendix A reports current-controller sensitivities and official-policy/code diagnostics. Within each endpoint all routes share one generation contract. The fixed- $N=45$ and prospective equal-cap BrowseComp routes use Qwen model-card settings with seed 20 and thinking disabled to keep decoding policy matched under the common caps; the post-hoc repeat changes only the seed to 21. The $N=80$ manifest and both judges use temperature 0, while GPT-5.6 rows omit unsupported temperature. Self-hosted equal-cap routes are pinned by model revision and container digest and hosted models by dated slug. Exact settings and identifiers are reported in Appendix H. Costs use uncached list prices and token logs; they exclude engineering effort and successful-completion totals can understate attempted scaffold billing.

Protocol and evidence tiers. Each headline claim specifies task, baseline and scorer. Controlled families establish mechanisms under identical generators; public Oolong and BrowseComp rows test transfer; LongBench-v2 diagnostics probe context limits, MCQ position, schema rigidity and split overlap. We label each public row as validation, ablation, replication or pilot under its stated contract. Appendix H records the model, protocol, compute and release boundaries needed to interpret the evidence.

Estimands, scoring and uncertainty. For paired routes r, s on n rows, $\hat{\Delta}_A = n^{-1} \sum_i (Y_{ir} - Y_{is})$ and $\hat{\Delta}_C = n^{-1} \sum_i (C_{ir} - C_{is})$; failed or invalid executions have $Y = 0$. Non-dominated routes form the observed task-conditional frontier. RLM is dominated when $\hat{\Delta}_A \leq 0$ and $\hat{\Delta}_C > 0$; positive tradeoffs require deployment-specific valuation. Paired accuracy uses McNemar tests (exact for small discordance); b, c denote first-only and second-only correct. Difference intervals use fixed-seed percentile bootstraps with 50,000 paired-row draws for BrowseComp and 5,000 context-window-cluster draws for Oolong. Primary controlled estimands are paired accuracy and captured cost

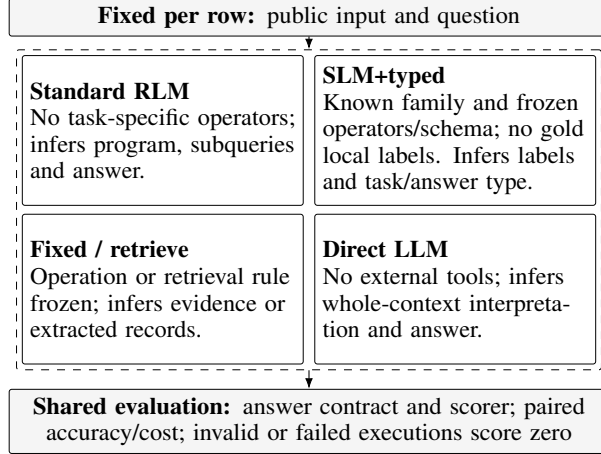


Figure 1: Route and information contracts. Allowed deployment prior knowledge is declared before comparison. The estimates compare complete routes rather than isolating a scaffold-only causal effect, and they do not evaluate an automatic router.

on T-MIXED and standard-unlabeled Oolong. BrowseComp-Plus supplies separate open-search estimates: the original $N=80$ manifest is bounded by two-episode completion and post-stop visibility; an attempted- $N=49$ endpoint stopped score-blind; an initial replication fixes $N=45$ before output; and the strongest prospective endpoint freezes 45 further untouched queries with iterative search and matched decision/time caps, following matched-budget practice [Alizadeh et al., 2026]. Failures are incorrect in every fixed denominator. Subgroup, controller and mechanism-probe tests are descriptive; we make no global familywise claim. The equal-cap endpoint applies Holm adjustment over RLM versus BM25, textual decomposition and iterative search. Oolong scores exact matches and accepted relative-frequency substrings, with $0.75^{|\Delta|}$ for numeric near-misses; we report both the binary exact rate and the partial-credit mean score. MCQ predictions are mapped back after answer shuffling. Because Oolong questions can share contexts, uncertainty resamples 72, 73, 32 and 20 clusters for $N=120$, SU-150, CD-45 and VC-45. Intervals are conditional on observed executions; a full CD-45 repeat documents rollout sensitivity for that frozen Oolong endpoint.

Cost accounting. RLM cost is decomposed into controller generation, child-query execution and final extraction. API totals capture successful completions and are lower bounds where failed calls return no usage. Direct and SLM baselines are single-call unless retrieval-augmented; fixed operators have zero marginal model cost but nonzero engineering cost. At SU-150, unrounded observed API savings are $\$0.022040733/\text{case}$ ($\$0.022041$ rounded), so a hypothetical $\$1,000$ implementation budget breaks even after $\lceil \$1,000/\$0.022040733 \rceil = 45,371$ cases [Kapoor et al., 2025]. This is an amortization sensitivity, not a labor estimate or TCO. Self-hosted BrowseComp costs use active runtime by route with generation-plus-judge total reported separately. Reported statistics and cost totals are computed from stored logs.

3 Results

Matched routes lead on known operations. On T-STRUCT, Gemma extracts records and a fixed Python operator performs the aggregation, which is exact by construction, so this row measures extraction success and cost. It reaches 100% accuracy versus 96.7% for RLM at about $99\times$ lower captured deployed-configuration API cost; because both are near ceiling, the result establishes cost dominance only, and supports neither a reliable accuracy difference nor a pure matched-model scaffold-overhead estimate. T-MIXED requires semantic judgments. A matched Gemini ablation separates finalization from semantic error: the count-only finalizer returns no valid answers while the JSON finalizer returns valid answers on all 20 examples and solves 13. A frozen $N=30$ endpoint compares direct Gemini with JSON-finalized RLM: direct Gemini solves 28/30 at $\$0.0223$, RLM solves 18/30 at $\$0.1089$ (paired exact $b=12$, $c=2$, $p=0.0129$) and remains about $4.9\times$ costlier. We treat T-NLP only as an interface diagnostic because no official RLM accuracy run is logged; moreover,

Table 3: Main results and mechanism boundaries by evidence tier. LP entries are mean/exact; others are exact. “Best non-RLM” prioritizes accuracy on T-MIXED. T-STRUCT compares deployed configurations; CD-45 is diagnostic. Equal-cap BrowseComp route-attributed active-runtime cost is \$13.279 (\$15.015 with interrupted-episode lower-bound billing and judging).

Regime	Best non-RLM	Standard RLM	Boundary supported by this row
T-STRUCT exact aggregation	Gemma extract + fixed agg.: 100%, \$0.0002/ex	96.7%, \$0.0198/ex	Known exact aggregation: a typed pipeline matches/exceeds RLM at much lower captured deployed-configuration API cost (about 99 $\times$ in this cross-model run; still 9.9 $\times$ if Gemma extraction is repriced 10 $\times$ higher).
T-MIXED directional count	Direct Gemini: 28/30, \$0.0223 total	RLM: 18/30, \$0.1089 total	Same-model direct inference wins on the frozen $N=30$ endpoint; the $N=20$ finalizer ablation isolates answer-contract failure.
Oolong specialization stress (CD-45; 1k/4k/262k)	Correct-family SLM+typed: 36/45; wrong-family: 2/45	Same-tools RLM: 31/45 or 29/45; GPT-5/GPT-5.6 Luna: 23/45, 26/45; at least 2.3 $\times$ costlier	Correct-family knowledge is necessary; same tools or tested controller changes narrow but do not reverse the ordering.
Oolong label-provided LP-80/LP-150	Typed ops: 100% / 97.3%	RLM: 82.4% / 84.7%	RLM is strong under flexible output schemas but gold-label typed operators expose the schema ceiling.
Oolong standard-unlabeled (SU-150; VC-45)	SLM-labeler + typed ops: 98/150, \$0.839; unused-source VC-45 35/45, \$0.312	RLM 68/150, \$4.145; VC-45 28/45, \$1.139	Five 8k–131k buckets plus pre-output-locked source-corpus transfer: SLM+typed has the best accuracy/cost point; VC-45’s exact-rate interval includes zero while its mean-score interval does not.
BrowseComp prospective equal-cap ($N=45$)	Text: 37/45, \$1.400; iterative: 33/45, \$1.710; BM25: 25/45, \$0.296	RLM: 31/45, \$9.873	The earlier RLM–BM25 gain does not transport: no RLM contrast survives Holm correction, and RLM is 5.8–7.1 $\times$ costlier than the stronger adaptive routes.

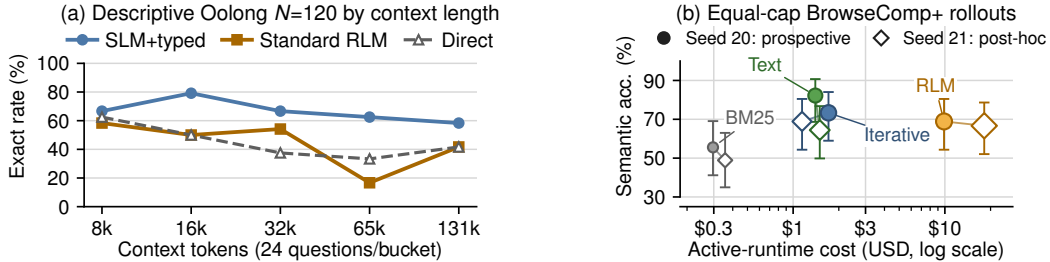


Figure 2: Accuracy and cost across task regimes. (a) Descriptive exact rates on standard-input Oolong development data ($N=120$; 24 per bucket). The inferential contrast is the paired cluster-resampled difference reported in the text, not a per-cell interval. (b) Accuracy versus active-runtime cost on the matched-Qwen equal-cap endpoint (filled circles: prospective seed 20; hollow diamonds: same-row post-hoc seed 21; $N=45$ each). Whiskers are Wilson intervals; marker area encodes all-row calls (RLM/iterative/text/BM25: 476/426/311/42 and 667/276/312/42). No RLM contrast survives Holm correction in either rollout.

146 the classification target requires categorical labels while the current aggregation scaffold assumes
147 numeric-style answers.

148 **Open-search advantages do not survive equal caps.** BrowseComp-Plus provides natural open
149 search without a known typed operator and already reports retriever/agent separation and cost
150 analyses; we add operation-matched routes under equal caps on frozen shards. The original frozen
151 $N=80$ manifest uses 1,000 documents per query and matched-Qwen routes: standard RLM, BM25
152 and textual decomposition. It was completed across two disclosed episodes after ranks 1–37 became
153 score-visible. A method-blinded Gemma 4 judge, sharing no model family with any judged route,
154 scores semantic equivalence to the benchmark’s gold answer and gives RLM 54/80 (67.5%), text

155 48/80 (60.0%) and BM25 39/80 (48.8%); only the 18.8-point RLM–BM25 interval excludes zero.
156 A separate GPT-5.6 blind audit likewise supports RLM over BM25 while leaving the RLM–text
157 comparison uncertain.

158 In a fixed- N replication, we froze 45 rows from a different official query shard with zero overlap
159 against earlier manifests. Model, routes, scorer and failure accounting were unchanged. Gemma 4
160 scores RLM 33/45 (73.3%), text 29/45 (64.4%) and BM25 21/45 (46.7%). RLM exceeds BM25 by
161 26.7 points (paired 95% interval 11.1–44.4; Holm $p=0.0151$), while its 8.9-point lead over text is
162 uncertain (−6.7–24.4). A blinded GPT-5.6 Sol audit agrees on 128/129 successful outputs ($\kappa=0.983$).
163 RLM completes 42/45, costs \$6.746 and has a 425.7-second p95 latency, versus 45/45, \$1.610 and
164 44.9 seconds for text.

165 The prospective equal-cap endpoint changes this ordering. We froze 45 untouched queries, added
166 iterative lexical search and attempted all 180 Qwen3.6 route cells under common 20-decision/1,200-
167 second caps before semantic scoring. A rank-19 interruption was recovered under the frozen
168 hash-bound pre-score protocol. RLM scores 31/45 (68.9%), versus iterative 33/45 (73.3%), text 37/45
169 (82.2%) and BM25 25/45 (55.6%). Differences are −4.4, −13.3 and +13.3 points, with discordances
170 6/8, 5/11 and 10/4. Paired 95% intervals are [−20.0, 11.1], [−31.1, 4.4] and [−2.2, 28.9] points; no
171 contrast survives Holm correction ($p_{\text{Holm}} \geq 0.539$). Sol/primary agreement by route is 41/41, 45/45,
172 45/45 and 41/42 (RLM/iterative/text/BM25; $\kappa=0.985$).

173 RLM completes 41/45 with 50.1/628.8-second median/p95 latency. Iterative/text complete all 45
174 with 14.1/90.2 and 22.1/47.6 seconds; BM25 completes 42/45. On 39 common completions, scores
175 are 30/28/31/22; this post-treatment sensitivity leaves the adaptive routes near parity. A question-only
176 Qwen leakage probe scores 0/45 under both judges: neither finds a correct answer without the corpus.
177 The earlier RLM–BM25 advantage therefore does not transport; Figure 2(b) plots both equal-cap
178 rollouts against active-runtime cost. Open search alone is not a sufficient routing signal.

179 A post-hoc seed-21 repeat reaches the same inferential conclusion. RLM/iterative/text/BM25 com-
180 plete 37/45, 45/45, 45/45 and 42/45 and score 30/31/29/22; no RLM contrast survives Holm
181 ($p_{\text{Holm}} \geq 0.346$). A cap-valid completion finishes within 1,200 seconds: RLM has 36/45, and
182 on 34 rows cap-valid for every route the scores are 28/23/20/18. This post-treatment sensitivity puts
183 RLM first, separating attrition from conditional accuracy. Across four descriptive cross-seed tests,
184 RLM is stable ($b=8, c=7, p=1.0$) while text shifts ($b=9, c=1, p=0.021$); the latter does not survive
185 four-test correction. RLM costs 12.1–15.9× the adaptive routes. Sol agrees on 166/169 outputs
186 ($\kappa=0.961$); the repeat is not pooled with the prospective endpoint.

187 **Oolong specialization persists across labels and sources.** On label-provided LP-80, RLM reaches
188 82.4% mean score against 70.0% for a generic fixed-parser/SLM fallback, although that difference
189 is uncertain ($p=0.089$); its largest gain is on timeline questions and deterministic parsing remains
190 perfect on counting. Gold-label typed operators reach 100% on the same split, so RLM is not
191 the strongest LP-80 route. On frozen held-out LP-150, RLM reaches 84.7% exact while typed
192 rules fixed before evaluation reach 97.3% ($b=23, c=4, p<0.001$). Gold local labels make these
193 aggregation/output-schema ablations, not end-to-end Oolong results. They show that RLM handles
194 flexible schemas better than a generic parser but not that open-ended Python is uniquely required
195 when the schema family is known.

196 All central standard-unlabeled checks use context_window_text without gold local labels. An
197 $N=120$ development replication established the SLM>RLM direction across every 8k–131k bucket
198 (Figure 2(a)), with a paired 22.5-point exact-rate advantage ($b=37, c=10, p=9.8 \times 10^{-5}$; 72-cluster
199 interval 11.9–33.0). The fixed pipeline was evaluated on SU-150, CD-45 and pre-output-locked
200 VC-45; before scoring, one parser repair was applied only to rows returning no call, a condition
201 visible without labels.

202 On SU-150, SLM+typed solves 98/150 at \$0.839, versus RLM 68/150 at \$4.145 and direct Gemini
203 65/150 at \$1.939. Discordance favors SLM over RLM ($b=44, c=14, p=1.00 \times 10^{-4}$); no direct–RLM
204 difference is detected ($p=0.788$). The 73-cluster interval gives a 10.6–29.9 point SLM advantage but
205 all rows reuse previously evaluated context windows so SU-150 is scale/method-lock evidence only.

206 VC-45 removes the known source-corpus overlap: its official validation rows come from spam
207 and trec_coarse, absent from the test split and all previous Oolong evaluations. The question
208 parser agrees with metadata on all 45 VC-45 rows; SU-150 and CD-45 carry no equivalent check.

SLM+typed solves 35/45 at \$0.312, versus RLM 28/45 at \$1.139 and direct Gemini 23/45 at \$0.658. SLM–RLM discordance is $b=10, c=3, p=0.092$ (Holm $p=0.185$); the 20-cluster exact-difference interval is 0.0–31.1 points while the mean-score interval is 3.0–30.9. RLM reaches 13/15 on user questions but 6/15 on counting. Task and source are not factorially balanced, so subgroup differences are descriptive. The specialized route remains cheaper and directionally favored under source-corpus transfer but VC-45 alone establishes neither a significant exact-rate advantage nor document-level independence.

CD-45 retains the specialization controls: wrong-family typing collapses from 36/45 to 2/45 while giving free-form standard RLM the same operator library reaches 31/45 or 29/45 but remains at least $2.3\times$ costlier. An independent standard-RLM repeat drops from 27/45 to 21/45. Together, SU-150 supplies scale, VC-45 supplies source-corpus transfer and CD-45 isolates specialization and same-tools behavior; none is a full Oolong leaderboard.

The Oolong ordering persists across tested controllers. The primary Oolong evaluation uses Gemini 2.5 Flash with the released standard-RLM stack and complete cost logs. Strict-stop GPT-5 and GPT-5.6 Luna standard-RLM runs on CD-45 reach 23/45 and 26/45; SLM+typed remains higher ($p=0.00235$ and $p=0.041$). On frozen CD-15, the official trained Qwen policy reaches 10/15 and the prospectively frozen official-code λ -RLM row reaches 8/15, versus 15/15 for SLM+typed. The complete λ row attempts all IDs on one pinned Qwen3-8B/vLLM deployment and counts two 1,800-second timeouts incorrect. Paired exact outcomes favor SLM+typed ($b=7, c=0, p=0.0156$); no difference is detected against direct Gemini, standard RLM/Gemini or official Qwen (all $p\geq 0.625$). Across controllers and policies, no sensitivity reverses the SLM+typed ordering; different manifests, denominators and fidelity tiers preclude pooling. These checks support controller robustness of the Oolong ordering within the evaluated contracts. The CD-15 diagnostics are not reproductions of the official SRLM/ λ -RLM/RAH benchmark protocols, so the central claim remains scoped to the released standard implementation.

The observed frontiers imply a task-conditional routing rule. Evidence supports two tiers. When task type is known or reliably inferred from the query, test the cheapest mechanism that matches the operation: fixed operators for exact aggregation, retrieval or semantic workers for matching, SLM/direct calls for local classification and question-parsed typed aggregation for known Oolong-style aggregation families. When no tested typed route captures the request, compare retrieval, textual decomposition and iterative search before treating RLM as a candidate. The Oolong results show that flexible schemas need not require open-ended generated Python; the prospective BrowseComp endpoint shows that open search does not itself justify RLM.

4 Mechanism Analysis

Observed failures span task interpretation, evidence selection, aggregation, tool use and finalization rather than a single RLM component. Appendix C.1 organizes these observations into known-operator overhead, final-answer contract sensitivity, semantic-local errors, output-schema rigidity, open-search instability, context-window constraints and multiple-choice format bias.

LLM+SLM helps with semantic decomposition, not fixed operations. The evaluation also tests the intuitive alternative "use a strong LLM to orchestrate cheap SLM workers" [Narayan et al., 2025]. This helps when the controller mainly decomposes semantic work but it is not an automatic replacement for deterministic operators. On exact tasks, a controller is unnecessary because the program is known. On T-NLP, LLM+SLM reaches 83.3%, compared with 87.5% for SLM direct while costing about $3.4\times$ more within the evaluated route set. On LB2-style QA, it inherits retrieval and MCQ-bias errors unless the evidence-selection stage is corrected. The viable version is therefore task-conditioned: use LLM+SLM for local interpretation, not for operations already solved by fixed code or retrieval.

Traces expose failures at distinct stages. Four reported CD-45 cases in Appendix C.2 cover a relation error, a direct whole-context error, a typed tie/normalization edge and same-tool interface overhead. Stored T-MIXED traces likewise show successful dual-gate filters but also vocabulary overreach and brittle finalization. These traces localize where failures surface; they do not attribute them to the scaffold as a whole.

Multiple-choice position bias limits the QA diagnostic. LongBench-v2 Academic QA is multiple-choice so answer-position priors confound retrieval-vs.-RLM comparisons. On the $N=50$ discovery set, canonical BM25 predicts A 68% of the time while gold-A is 18%. Rotating priming improves pooled $N=94$ ($p=0.015$) but not held-out $N=44$ ($p=1.0$); single-call accuracy is unmeasured. We therefore keep it as a validity diagnostic, with prior debiasing comparisons in Appendix B.

5 Related Work

RLM externalizes context through controller-generated programs and recursive calls [Zhang et al., 2025]. Follow-ups question recursion as the driver or add typed operators, coding agents and recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Ehrlich and Blackman, 2026, Cao et al., 2026, Lumer et al., 2026]. Wang finds task-dependent depth, cost and strong-model degradation [Wang, 2026]. Our operation-matched comparison fixes rows and scorers, reports paired uncertainty and treats the deployed route rather than recursion depth as the unit of analysis.

Cost-aware scaffold evaluation shows that simpler baselines can Pareto-dominate agents [Kapoor et al., 2025]. This study applies that principle to complete long-context routes under explicit information, scoring and budget contracts and measures recommendation stability across rollouts.

Oolong separates local labeling from aggregation [Bertsch et al., 2026]; its labels support our mechanism ablations. Long-context suites test realistic reasoning, effective context and retrieval/length effects [Bai et al., 2025, Chen et al., 2026b, Hsieh et al., 2024, Yen et al., 2025, Zhang et al., 2024, Modarressi et al., 2025, Li et al., 2025]. Decomposition, search and routing systems select evidence or execution routes [Xu et al., 2026, Song et al., 2026, Ong et al., 2025, Li et al., 2024, Chen et al., 2026a]. Corpus-scale evidence shows that lexical and agentic search can reverse order as the search space grows [Wang et al., 2026]. We compare fixed, worker and RLM routes under logged cost and explicit evidence grades.

6 Limitations

External validity. The study diagnoses task-conditional failure modes, not average long-context performance. VC-45 is source-corpus but not document disjoint because Oolong lacks IDs; SU-150 reuses windows. Official-family evidence covers CD-15. The controlled curve ends at 131k; longer rows are stress tests.

Measurement. Equal-cap evaluation attempts all 180 cells under fixed caps across three execution episodes. Evaluation validity is assessed through cross-family judge agreement and a question-only leakage control. The study does not include human adjudication, so these checks establish stability rather than human validity amid contextual judge fragility [Tan et al., 2025, Xu et al., 2025].

Cost and scope. Costs exclude engineering and TCO; alternatives are not exhaustive and recursion depth is fixed. BrowseComp bins reach $n=10$, so we infer no subgroup effects. A same-row embedding rerank remains below RLM but is not full-corpus dense retrieval (Appendix B). Seven repeat RLM calls exceed the cap and score incorrect. Self-hosted and API dollars remain separate. Automated route choice could suppress useful methods in untested domains.

Ethics and AI assistance. Public or synthetic benchmarks involve no human subjects. AI tools assisted with coding, experiment orchestration and language editing; the authors verified the analyses and manuscript.

7 Conclusion

Matched routes lead on known operations. Across two equal-cap rollouts no RLM contrast survives multiplicity correction while RLM costs $5.8\text{--}15.9\times$ more. RLM evaluation should therefore report a task-conditional route frontier, not promote a single scaffold ranking into a deployment policy.

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431 A Protocol Details

432 **Cost-aware audit protocol.** Table 4 summarizes the protocol used throughout this study. Because 433 the audited object is a scaffolded inference procedure, a row should not be promoted from diagnostic 434 evidence to method evidence unless the input contract, output schema, scorer, failure accounting and 435 cost trace are matched.

436 **Role-based model coverage.** The primary panel assigns models to deployment roles rather than 437 constructing a broad leaderboard: Gemini 2.5 Flash controls the logged standard RLM and supplies 438 direct frontier inference, Gemma 4 26B-A4B supplies SLM worker paths and GPT-4o-mini controls 439 the short T-NLP LLM+SLM route. GPT-5 is used for the T-STRUCT frontier-controller exception 440 and a CD-45 controller sensitivity. Many central alternatives are deterministic or structural so the 441 unresolved breadth question is same-contract coverage of stronger RLM-family systems rather than 442 additional direct-LLM columns.

443 That stronger-family evidence includes frozen-manifest adapter pressure tests, matched standard- 444 unlabeled smokes, the CD-15 repeatability study and CD-45 same-operator-library diagnostics. 445 The unmodified public λ -RLM repository attempts all 15 frozen CD-15 IDs with its published 446 Qwen3-8B settings and reaches 8/15 exact with two declared timeouts. MIT OASYS’s official 447 `r1m-qwen3-30b-a3b-v0.1` post-trained policy reaches 10/15 exact with one timeout through the 448 canonical `r1m` harness and model-card Oolong settings. The λ row is official-code cross-task evidence 449 rather than an original-paper reproduction; rankable official SRLM and same-contract RAH rows 450 remain unavailable.

451 Controller-freshness and compatibility checks are reported separately from family parity. Table 5 452 gives the contracts for the complete central and sensitivity rows; these checks do not constitute full 453 official SRLM/ λ -RLM/RAH benchmark parity.

Step	Check	Audit rule
1	Scope	Name the scaffold, input field, answer contract, scorer and non-claims before comparing methods.
2	Match rows	Run the scaffold and alternatives on identical instances; separate development, fixed-pipeline and split-validity evidence.
3	Cheap route	Include deterministic code, retrieval/reranking, SLM workers, direct LLM and RLM only where each is natural for the operation.
4	Cost/failures	Record per-example cost, timeout/error rows, retry/stop rules and attempted-cost lower bounds.
5	Confounds	Add finalizer, schema, same-operator, wrong-family, repeatability and trace probes when results could be harness artifacts.
6	Claims	Mark rows as confirmed, diagnostic, smoke, proxy or future; report falsifiers and candidate-positive regimes separately.

Table 4: Cost-aware audit protocol used in this study. Standard RLM is compared only after cheaper operation-matched mechanisms receive the same instances, answer contract, scorer, cost logging and failure accounting.

Slice / row	RLM controller	Row cap	Interpretation
Oolong CD-45 central	Gemini 2.5 Flash	360s	Main standard-rlms row.
Oolong CD-15 repeats	Gemini 2.5 Flash	360s	Repeatability diagnostic over a frozen subset.
Oolong CD-15 official trained policy	Qwen3-30B-A3B + official LoRA	1500s	Canonical-harness, same-contract diagnostic; one timeout is scored incorrect.
Oolong CD-45 same-operator RLM	Gemini 2.5 Flash	360s	Standard-rlms custom-tool diagnostic, not official SRLM/ λ -RLM.
Oolong CD-45 controller sensitivity	GPT-5	240s	Controller+stability sensitivity, not pure controller ablation.
Oolong CD-45 GPT-5.6 complete	GPT-5.6 Luna	360s	Current-controller sensitivity; complete strict-stop row, not official SRLM/ λ -RLM/RAH.
Oolong CD-15 official λ code	Qwen3-8B	1800s	Complete unmodified-repository cross-task row; 15/15 attempted, 13 completed, 8 exact.
LB2 >500k smoke	Gemini 2.5 Flash	480s	Stop-rule feasibility smoke, not stable ranking.

Table 5: Protocol caps for complete claim-central and sensitivity rows. The rows answer different questions and should not be merged as pure model swaps.

454 **Cost accounting.** RLM cost aggregates controller program generation, recursive or child-query
455 execution and final answer extraction. Direct LLM/SLM baselines are single-call unless retrieval-
456 augmented. Fixed operators have zero marginal LLM cost but not zero engineering cost; hybrid
457 pipelines such as T-STRUCT include their SLM extraction calls. Current RLM cost averages are
458 successful-completion costs from row-level logged prediction/cost/error files; not every central
459 RLM row retains a full raw generated-program/provider-payload trajectory. The release includes
460 an attempted-cost audit script and JSON artifact that scan local `model_calls*.jsonl` files for
461 official-RLM completion rows rather than any one headline run. Across those 85 logged attempt rows,
462 76 completed rows record \$1.289 total cost while 9 failed timeout/error rows record zero captured
463 usage/cost. Thus the local logs support a recorded attempted-cost lower bound across archived

Method	Exact	Acc.	Cost	Invalid ex.
BM25 + Gemma reader	8/30	26.7%	\$0.0355	3
BM25 + Gemini reader	11/30	36.7%	\$0.1588	2
BM25 + Gemma rerank + Gemini	12/30	40.0%	\$0.1863	2

Table 6: Balanced LongBench-v2 train-split >500k-character retrieval/rerank diagnostic ($N = 30$, five examples from each domain, seed 20260714). All rows use local BM25 over the full context and three answer-order votes. The rerank row applies Gemma to score 24 BM25 candidates before Gemini reads the selected excerpts. This is an over-window alternative diagnostic, not a same-row RLM-family comparison.

Method	Exact	Acc.	Cost	Errors
Standard r1ms	1/6	16.7%	\$0.4698	1
BM25 + Gemma reader	1/6	16.7%	\$0.0065	0
BM25 + Gemini reader	2/6	33.3%	\$0.0284	0
BM25 + Gemma rerank + Gemini	2/6	33.3%	\$0.0326	0

Table 7: Same-row standard-r1ms smoke on the balanced LongBench-v2 train-split >500k-character manifest. The six rows are the shortest eligible context in each balanced domain. This is a stop-rule feasibility check, not a full $N = 30$ ranking.

464 official-RLM model-call traces but they cannot reconcile to a single Oolong/T-MIXED headline
465 denominator or reconstruct provider-side billing for failed partial scaffold calls whose usage was not
466 returned.

467 Because person-hour estimates are unavailable, we report a deterministic audit of implementation
468 surface. The complete released typed-route package spans four source files and 2,017 physical lines
469 (1,793 nonblank) including the SLM labeler, typed aggregation kernel, natural-language schema
470 parser, standard-RLM tool wrappers, command-line interfaces, caching, tracing and analysis sup-
471 port. The release declares schemas for six families and five answer types. Three families appear
472 as controlled families in the main paper; the others support supplement-only diagnostics or public
473 anchoring. The standard-RLM diagnostic exposes five atomic/schema tools and one optional conve-
474 nience solver. These counts are an auditable upper bound on the released implementation surface, not
475 a minimal runtime kernel or a measure of human engineering time, maintenance burden or total cost
476 of ownership.

477 We also parameterize the unmeasured one-time implementation cost instead of treating it as zero. On
478 SU-150, the observed successful-completion API-cost difference is \$0.022041 per case; amortizing
479 assumed one-time budgets of \$100, \$1,000 and \$10,000 from API savings alone requires 4,538,
480 45,371 and 453,704 cases when the ratio is computed from unrounded costs and rounded up to
481 the next whole case. On VC-45, the corresponding difference is \$0.018377 and the thresholds are
482 5,442, 54,416 and 544,158 cases under the same rule. These are deployment-volume sensitivities,
483 not estimates of actual labor or maintenance. They omit failed-call billing, latency value, reliability
484 and future model-price changes; SU-150 also reuses prior windows while VC-45 is source-corpus
485 transfer without document-level independence.

486 A deterministic cost-sensitivity audit reconciles the stored result files. T-STRUCT’s roughly $99\times$ gap
487 is a deployed-configuration comparison: GPT-5-backed standard RLM costs \$0.019826/example on
488 the stored run while the Gemma extraction plus fixed aggregation route is estimated at \$0.0002/exam-
489 ple. It is not a pure matched-model estimate of scaffold overhead. For the Oolong $N = 30$ pilot, the
490 corrected SLM-labeler token-trace price is \$0.007257 versus \$0.255676 for RLM ($35.2\times$); the older
491 pre-patch SLM estimate was \$0.103732 and still leaves RLM $2.46\times$ more expensive. On the Oolong
492 $N = 120$ public-contract run, RLM costs \$2.688340 versus \$0.658995 for SLM+typed ($4.08\times$).
493 The larger $N = 150$ fixed-pipeline replication gives \$4.144901 versus \$0.838777 ($4.94\times$ captured
494 successful-completion cost). Exact prices may change, but the reported cost comparisons survive the
495 disclosed sensitivity checks within their stated deployment scope.

496 B Diagnostic Validity Controls

The balanced diagnostic covers Code Repository Understanding, Long In-context Learning, Long Structured Data Understanding, Long-dialogue History Understanding, Multi-Document QA and Single-Document QA. The strongest row, BM25+Gemma-rerank+Gemini, reaches 3/5 on code, 2/5 on in-context learning, 2/5 on structured data, 1/5 on dialogue, 3/5 on multi-document QA and 1/5 on single-document QA. Reranking changes three rows and improves only one net example over BM25+Gemini. We therefore use the result to calibrate the over-window boundary: cheap retrieval/reranking is a necessary comparison for RLM but this particular route does not close the extreme-context region.

Multiple-choice position control. On the LongBench-v2 Academic QA discovery set ($N = 50$), canonical BM25 predicts option A on 68% of rows while only 18% of gold answers are A. Rotating answer-order priming improves the pooled $N = 94$ comparison ($p = 0.015$) but not the held-out $N = 44$ slice alone ($p = 1.0$). Single-call accuracy was not separately measured, so this remains a format-validity diagnostic rather than evidence for a retrieval or RLM ranking.

Post-hoc dense-rerank sensitivity. On the attempted- $N = 49$ BrowseComp-Plus replication rows, a same-row post-hoc route reranks 64 BM25 candidates with Qwen3-Embedding-8B before the same Qwen answer model and Gemma judge. The hybrid scores 20/49 at \$0.516 versus 22/49 at \$0.509 for matched BM25 replay and 35/49 for RLM; RLM-hybrid is +30.6 points (interval 14.3–46.9; $p = 0.0026$). Required-document recall rises only from 0.256 to 0.261. This is not full-corpus dense retrieval and all recall diagnostics were computed after output.

C Mechanism and Trace Analysis

C.1 Mechanism Inventory

Table 8: Mechanism-level error analysis grouped by layer, setting and deployment implication.

Mechanism	Where observed	What fails	Paper implication
Known exact operator	T-STRUCT, Oolong counting	Programs add calls when the symbolic operation is known.	Typed extraction plus deterministic code dominates standard RLM at the observed operating points.
Final-answer contract	T-MIXED, T-NLP, official λ row	Valid content is not finalized under the answer contract.	Evidence localizes a finalization layer but not the whole scaffold.
Semantic local judgment	T-MIXED	Generated lexical programs miss directional meaning.	Direct local judgment beats repaired RLM in this family.
Output-schema rigidity	Oolong timeline/user	A generic parser assumes numeric outputs where labels or user IDs are required.	Typed operators remove this failure only when the schema family is inferable.
Open search	BrowseComp-Plus	Descriptive route ordering changes across equal-cap rollouts.	Open search alone does not select RLM; matched alternatives, completion and cost tails matter.
Context-window constraint	LB2 >500k	Direct input is infeasible and retrieval can miss distributed evidence.	Over-window inputs motivate decomposition, not an untested RLM benefit claim.
Evaluation-format bias	LB2 multiple choice	Retrieval outputs are sensitive to answer-position priors.	Debiasing is required before interpreting retrieval–RLM differences.

C.2 Representative Trace Evidence

Table 9: Representative released CD-45 traces selected by outcome pattern. These are illustrations; aggregate conclusions remain tied to the full frozen rows.

Case	Observed behavior	Localized failure	What it establishes
912090010	Gold is “less common.” SLM+typed, direct Gemini and GPT-5 RLM are correct; standard RLM/Gemini returns “same frequency.”	Relation/count error	Direct and structured routes recover an error made by one scaffold roll-out.
312030020	Gold user is 16226. Both RLM controllers and SLM+typed recover it; direct Gemini returns 72294.	Direct whole-context aggregation error	RLM can be useful on an individual case even when it is not the strongest aggregate route.
710070026	Gold least label is neutral. Standard RLM and direct Gemini are correct; SLM+typed returns contradiction despite 7/7 correct local labels.	Typed tie/normalization edge	Typed pipelines also fail above the local-label layer.
410040043	Convenience same-tools RLM succeeds after 9 iterations, 7 code blocks and 3 tool calls; atomic RLM uses 16 iterations, 15 blocks and 14 calls but emits a non-matching final form.	Tool-interface confusion and finalization	Shared operators can narrow accuracy differences while retaining orchestration overhead.

Group	N	Oracle typed	RLM	fp+SLM	SLM direct
Overall	80	100.0%	82.4%	70.0%	53.8%
Counting	29	100.0%	85.3%	100.0%	65.5%
Timeline	29	100.0%	79.9%	44.8%	37.9%
User	22	100.0%	81.8%	63.6%	59.1%

Table 10: Same-slice label-provided Oolong-synth aggregation ablation ($N = 80$, seed 42, max context 8192 tokens). All methods receive `context_window_text_with_labels` so gold local labels are visible; this is not standard unlabeled Oolong classification-plus-aggregation. RLM/fp+SLM/SLM rows are logged methods under this input contract; typed proxy is a post-hoc deterministic typed-combinator diagnostic with no model/API calls that also uses benchmark task/output metadata. Entries are mean Oolong scores: exact match for categorical/date-style answers, accepted substrings for relative-frequency comparison strings and $0.75^{|\Delta|}$ partial credit for numeric near-misses.

D Label-Provided Oolong Aggregation Ablations

The subgroup pattern isolates a mechanism under the label-provided aggregation contract. Counting favors deterministic parsing, whereas timeline/user identification favors RLM over the generic fp+SLM fallback because the fallback does not fix the required output type correctly in advance. The typed proxy localizes this gap to output-schema and harness design: once the Oolong task family and output metadata are expressed as bounded typed operators, the same slice is solved at 100%. This does not show that open-ended generated Python is necessary when labels are already supplied. The benchmark scorer gives three RLM examples fractional credit. Thresholding those scores at 1.0 leaves the ordering unchanged: RLM is 64/80 (80.0%) strict exact, versus 56/80 (70.0%) for fp+SLM, 43/80 (53.8%) for SLM direct and 37/80 (46.2%) for SLM+CoT. Both the RLM timeline advantage and the typed-proxy repair remain hypothesis-generating diagnostics because the subgroup tests are small, the input is label-provided and the proxy was added after failure inspection.

On exact outcomes over the frozen label-provided $N = 150$, typed v1 beats RLM with discordance $b = 23, c = 4$ (two-sided exact $p < 0.001$); the repaired typed row beats RLM with $b = 23, c = 0$ ($p < 10^{-6}$). RLM remains substantially stronger than the pure fixed parser ($b = 72, c = 11, p < 10^{-10}$). The public λ -RLM implementation pressure test scores 43.1% mean score / 42.0% exact, below standard RLM ($b = 73, c = 9, p < 10^{-12}$) and typed v1 ($b = 86, c = 3, p < 10^{-20}$) and statistically similar to the pure fixed parser on exact outcomes ($b = 27, c = 30$, n.s.). The prehend SRLM-style pressure test is much stronger than the λ adapter on the matched $N = 60$ subset (83.0% mean score; $b = 30, c = 3, p = 1.4 \times 10^{-6}$) and is not significantly different from standard RLM on paired exact outcomes ($b = 4, c = 10, p = 0.180$) but it remains below typed v1 ($b = 0, c = 10, p = 0.002$). Its failures are wrong numeric counts (4), wrong relations (3), wrong user IDs (3) and wrong labels (2). Thus the result is not that RLM-family search is weak but that under a gold-label

Method	N	Overall	Counting	Timeline/User
Fixed parser	150	44.0%	100.0%	14.0/18.0%
λ -RLM public impl.	150	43.1%	63.2%	32.0/34.0%
prehend SRLM-style	60	83.0%	90.3%	83.8/75.0%
Oracle typed v1	150	97.3%	100.0%	96.0/96.0%
Oracle typed repair	150	100.0%	100.0%	100.0/100.0%
RLM	150	86.9%	89.5%	79.1/92.0%

Table 11: LP-150: frozen held-out label-provided Oolong aggregation ablation (seed 20260712, no overlap with the same-slice LP-80). All rows use `context_window_text_with_labels`; standard unlabeled Oolong is not evaluated here. The v1 typed rules were fixed before new model calls; the repaired row adds two bounded operators after inspecting four misses. The RLM row uses the same OpenRouter Gemini 2.5 Flash `rlms` configuration as the same-slice run. The λ -RLM row uses the public implementation through a project-side frozen-manifest adapter, default 100k-character context window. The prehend SRLM-style row uses the public fork’s $K = 3$ self-consistency/confidence/trace-length selection on the first balanced $N = 60$ frozen examples. Entries are mean Oolong scores under this ablation.

aggregation contract a task-family typed harness is a better comparator than a generic parser fallback or generic recursive/reflective QA adapter.

E Standard-Unlabeled Oolong Evaluations

$N = 120$: uncertainty and direct control. This public-contract replication covers five context lengths and 72 context clusters. Paired exact outcomes favor stored SLM-labeler+typed over standard RLM ($b = 37, c = 10, p = 9.85 \times 10^{-5}$). The N120 three-way controller JSON (full path in the manifest) gives context-cluster intervals of 66.7% [58.2, 75.0] for SLM+typed and 44.2% [35.6, 53.0] for RLM; paired differences are 11.9–33.0 exact-rate points and 14.1–33.7 mean-score points. A post-run no-call schema audit infers task/output type from each question and preserves 80/120 but reuses stored SLM labels rather than rerunning a prospectively locked parser. Direct Gemini, using the RLM controller model without recursion or code execution, reaches 54/120 exact and 50.0% mean score at \$1.546563. It is not detectably different from RLM ($b = 26, c = 25, p = 1.0$; cluster exact-difference interval -10.9–13.0) while SLM+typed still wins ($b = 30, c = 4, p = 6.16 \times 10^{-6}$; 12.4–30.3). This is therefore a deployed-stack comparison, not a scaffold-only causal estimate. Counting/timeline/user exact counts are 20/31/29 for SLM+typed, 15/16/23 for direct Gemini and 9/19/25 for RLM.

SU-150: scale and overlap. SU-150 is the largest standard-unlabeled Oolong tier by nominal N ; VC-45 remains the cleanest source-corpus transfer. The parser, typed operators, SLM prompt, scorer and cost rules were fixed before calls. A no-call question-parser audit agrees with metadata on 150/150 rows and reproduces the 98/150 SLM+typed result from stored labels. Paired exact outcomes favor SLM+typed over RLM ($b = 44, c = 14, p = 1.00 \times 10^{-4}$) and direct Gemini ($b = 42, c = 9, p = 3.39 \times 10^{-6}$); direct and RLM do not differ detectably ($b = 26, c = 29, p = 0.788$). Context-cluster intervals give SLM+typed 65.3% [57.3, 73.3] and RLM 45.3% [36.9, 54.0], with a paired advantage of 10.6–29.9 points. These intervals describe evaluated windows, not independence from development contexts: the split audit finds 20 prior ID overlaps and prior-window overlap for all 150 rows. Counting/timeline/user exact counts are 23/31/44 for SLM+typed, 18/19/28 for direct Gemini and 14/23/31 for RLM. SLM+typed also has the highest mean score in every length bucket (79.8, 67.8, 78.0, 71.8 and 59.5% from 8k to 131k); RLM is closest at 16k but has six timeout/error rows and the highest cost per exact answer. SU-150 is a larger known-family fixed-pipeline replication, not a context-disjoint holdout or open-domain leaderboard.

VC-45 was frozen before paid outputs after an exact-schema audit of the pinned Oolong snapshot. The freezer verifies zero overlap with the test split and prior project artifacts on row ID, context-window ID and upstream corpus. A no-call preflight exposed one narrow parser grammar omission for “spam or ham (i.e., not spam)”; admitting the parenthesis before any paid output gave 45/45 parser-valid rows. The post-run question-only parser agrees with benchmark task and answer metadata on all 45 rows and exactly reproduces the 35/45 typed result.

Method	Exact	Mean	Cost	Err.	\$/exact
Q-parsed SLM+typed	80/120	72.7%	\$0.659	0	\$0.0082
Direct Gemini	54/120	50.0%	\$1.547	0	\$0.0286
RLM	53/120	48.9%	\$2.688	9	\$0.0507

Table 12: Standard-unlabeled Oolong-synth multi-length result on a frozen balanced manifest ($N = 120$, three task groups $\times$ five context lengths $\times$ eight examples; 72 context clusters, max two questions per context). All rows use `context_window_text`; gold local labels are not present. The stored SLM-labeler + typed row is paired with a post-run no-call question-inferred schema audit within the known Oolong task family: it infers task/output schema from the natural-language question before aggregation and gives 100% agreement with benchmark metadata on these stored labels. This is deployability evidence, not a fully prospective open-taxonomy parser validation. Costs are corrected/logged API costs from traced calls.

Method	Exact	Mean	Cost	Err.	\$/exact
Q-parsed SLM+typed	98/150	71.4%	\$0.839	0	\$0.0086
Direct Gemini	65/150	47.8%	\$1.939	0	\$0.0298
Standard RLM	68/150	48.4%	\$4.145	6	\$0.0610

Table 13: SU-150: larger standard-unlabeled Oolong fixed-pipeline replication after freezing the question parser, typed operators, SLM prompt, scorer and cost rules. The manifest balances three task groups across five context lengths (10 examples per group/length cell; 73 context clusters; max context reuse 3). A split-overlap audit finds that it is not context-window-disjoint from earlier Oolong work. All rows use `context_window_text`; gold local labels are not present.

Method	Exact	Mean	Cost	Err.	\$/exact
Question-parsed SLM+typed	35/45	82.1%	\$0.312	0	\$0.0089
Direct Gemini	23/45	55.2%	\$0.658	0	\$0.0286
Standard RLM	28/45	65.6%	\$1.139	0	\$0.0407

Table 14: VC-45: pre-output-locked standard-unlabeled Oolong replication after one outcome-blind no-call parser repair. Its `spam` and `trec_coarse` source corpora do not occur in the official test split or prior project artifacts. Task and length margins are balanced but not every task-source cell; the official schema exposes no source-document identifier.

578 SLM+typed has the highest point estimate and the lowest captured successful-completion cost.
579 Against standard RLM, paired exact discordance is $b = 10, c = 3$ (two-sided exact $p = 0.0923$;
580 three-pair Holm $p = 0.1846$); the 20-cluster exact-rate difference interval is 0.0–31.1 points while the
581 mean-score difference interval is 3.0–30.9 points. SLM+typed beats direct Gemini with $b = 14, c = 2$
582 ($p = 0.00418$; Holm $p = 0.0125$). No direct–RLM difference is detected ($b = 8, c = 13, p = 0.383$).
583 By group, SLM+typed / RLM / direct exact counts are 10/6/9 for counting, 14/9/8 for timeline and
584 11/13/6 for user. Timeline rows all come from `spam` while counting and user rows span both corpora;
585 subgroup differences are therefore descriptive, not source-independent. Captured retained-row cost is
586 \$2.109802; the OpenRouter account delta is \$2.304960 because an aborted duplicate-worker launch
587 and partial in-flight calls add \$0.195158. All 45 SLM prompts/responses, direct prompts/responses
588 and standard-RLM input contexts, raw answers, 205 generated code blocks and trajectories are
589 retained.

590 **CD-45: split validity.** CD-45 is the cleanest project-history split check: after scanning prior Oolong
591 result artifacts, its manifest has zero overlap on example ID and context-window ID. The question
592 parser agrees with task/answer metadata on 45/45 rows and the same typed operators aggregate stored
593 SLM labels. Exact outcomes favor SLM+typed over RLM ($b = 11, c = 2, p = 0.0225$) and direct
594 Gemini ($b = 16, c = 1, p = 2.75 \times 10^{-4}$); direct is directionally below RLM ($b = 2, c = 8, p =$
595 0.109). Context-cluster intervals are 80.0% [68.6, 91.1] for SLM+typed, 60.0% [45.5, 74.4] for RLM
596 and 46.7% [30.4, 63.4] for direct; SLM–RLM intervals are 6.0–35.6 exact-rate points and 7.4–36.1
597 mean-score points. Counting/timeline/user exact counts are 12/11/13 for SLM+typed, 6/8/13 for
598 RLM and 8/5/8 for direct. At 1k/4k/262k, SLM+typed scores 13/11/12; direct falls to 4/15 at 262k

Method	Exact	Mean	Cost	Err.	\$/exact
Q-parsed SLM+typed	36/45	83.7%	\$0.416	0	\$0.0116
Direct Gemini	21/45	54.9%	\$1.025	0	\$0.0488
Standard RLM	27/45	62.3%	\$1.001	0	\$0.0371
Standard RLM + same typed ops	31/45	69.3%	\$1.314	2	\$0.0424

Table 15: CD-45: project-history context-window-ID-disjoint standard-unlabeled Oolong validation after excluding all previously used example IDs and context-window IDs found in archived artifacts. The manifest balances counting, timeline and user tasks across 1k, 4k and 262k windows, has 32 context clusters and permits at most two questions per window. Gold local labels are absent. This is not proven source-document-disjoint and its length mix differs from SU-150. Bootstrap intervals resample context-window clusters conditional on the observed method executions; a separate RLM repeat probes rollout sensitivity. Costs are captured successful-completion totals and are lower bounds for rows with failures.

while RLM remains intermediate. Oolong exposes no stronger source-document key and CD-45’s 1k/4k/262k mix differs from SU-150’s 8k–131k range. CD-45 therefore supports split validity under a shifted length mix, not document disjointness or a pure overlap ablation of SU-150.

A no-call selection audit makes that last caveat explicit. SU-150 has 150 rows over 8k/16k/32k/65k/131k windows and 73 context clusters but its archival split audit finds prior context-window overlap. CD-45 has zero prior project example-ID and context-window-ID overlap but uses 1k/4k/262k windows and all three methods score higher: SLM+typed 36/45 versus 98/150 on SU-150, standard RLM 27/45 versus 68/150 and direct Gemini 21/45 versus 65/150. The released JSON/MD selection audit supports the following interpretation: SU-150 supplies larger fixed-pipeline scale and method-lock evidence while CD-45 supplies cleaner split-validity evidence under a different length mix.

A separate no-call specialization stress test asks whether SLM+typed benefits primarily from selecting the correct operation family. It reuses the same stored CD-45 Gemma row-label predictions and changes only that family. The metadata route and natural-language question-parsed route both solve 36/45. An intentionally wrong-family route solves 2/45 and a naive generic most-frequent-label route solves 3/45. These stress routes are not competing methods; they show that the SLM+typed advantage depends on choosing the correct operation family. This supports the paper’s deployment-specialization framing and would have weakened the thesis if the wrong-family route had remained strong.

We also ran a paid shared-operator controller diagnostic on the same CD-45 stored SLM-label predictions. Gemini 2.5 Flash received only the natural-language question, the allowed typed Oolong operator schema and the answer-type schema. It selected the task and answer type and the same typed operators then executed deterministically over the stored row labels. This diagnostic has 45/45 task agreement, 45/45 answer-type agreement, 36/45 exact, 83.7% mean score, zero parse/execution errors and \$0.00886 captured controller cost. It reduces the concern that the typed route depends on hidden metadata routing but it is not standard `rlms`, not an official SRLM/ λ -RLM comparator and not evidence that a generic open-ended RLM scaffold would choose the same bounded operator program.

The direct same-operator-library standard-`rlms` diagnostic addresses that last point. We injected the same typed Oolong operator library through `custom_tools`, first including a high-level convenience solver and then withholding it in an atomic-only ablation while preserving standard `rlms` free-form program generation, standard `context_window_text` input and the natural-language question as the root prompt. Neither row is the shared-operator controller or official SRLM/ λ -RLM. On CD-45 the convenience row solves 31/45 exact (68.9%, mean score 69.3%) at \$1.314, with two 360-second timeouts, 173 tool calls and 348 generated code blocks. By group it is 11/15 on counting, 9/15 on timeline and 11/15 on user; by length it is 12/15 at 1k, 10/15 at 4k and 9/15 at 262k. Paired exact outcomes still directionally favor SLM+typed/shared-operator over same-operator RLM (SLM-only 7, same-operator-only 2, $p = 0.180$) while same-operator RLM is directionally above direct Gemini (direct-only 6, same-operator-only 16, $p = 0.052$) and not significantly different from the original standard-RLM rollout (original-only 8, same-operator-only 12, $p = 0.503$).

Case	Route outcomes	What it shows
RLM tie error 912090010, C/4k	Gemini RLM: same frequency (0); GPT-5 RLM, SLM+typed and direct: less common (1).	Fixed comparison repairs an RLM tie after accurate SLM row labels.
RLM label error + timeout 412040009, C/4k	Gemini RLM: False (0); GPT-5 RLM: timeout (0); SLM+typed and direct: true (1).	SLM+typed and direct answer correctly; GPT-5 RLM hits the row cap.
Direct aggregation error 312030020, U/4k	Both RLMs and SLM+typed: 16226 (1); direct: 72294 (0).	Direct selects the wrong most-frequent user; structured routes recover it.
Typed edge miss 710070026, C/1k	Gemini RLM and direct: neutral (1); GPT-5 RLM: timeout (0); SLM+typed: contradiction (0).	A typed tie/normalization edge fails despite exact local labels.
Long-context timeout 218020026, C/262k	Gemini RLM, SLM+typed and direct: more common (1); GPT-5 RLM: timeout (0).	Three routes agree; GPT-5 RLM times out at 262k.

Table 16: Representative CD-45 paired examples from the no-call trace artifact. Parentheses mark exact correctness. The SLM+typed and direct-controller rows have stored prompts/raw responses and token/cost traces. Aggregate CD-45 standard-RLM rows retain predictions, costs and errors; a separate three-row trajectory subset captures generated code blocks for qualitative inspection so this table remains behavior-first rather than a new generated-program benchmark.

639 The atomic-only variant retains the schema-list tools and the three functions below but removes the
640 high-level `solve_oolong_with_tools` convenience solver:

```
641     parse_oolong_rows
642     label_oolong_rows_with_slm
643     solve_labeled_oolong
```

644 It solves 29/45 exact (64.4%, mean score 69.3%) at \$0.959, with four timeout/error rows, 155 tool
645 calls and 337 generated code blocks. By group it is 9/15 on counting, 8/15 on timeline and 12/15
646 on user; by length it is 11/15 at 1k, 11/15 at 4k and 7/15 at 262k. Paired exact outcomes favor
647 SLM+typed/shared-operator over atomic same-operator RLM (SLM-only 9, atomic-only 2) while
648 the convenience and atomic rows are close but not identical (convenience-only 8, atomic-only 6).
649 The failure trace contains wrong task/output choices, malformed final answers, self-reported tool-use
650 confusion, unsupported typed-solver calls and long-row timeouts. Supplying the operator library
651 therefore helps standard RLM, and the effect does not rest entirely on one convenience solver. Under
652 this bounded protocol, however, free-form orchestration still does not match the cheaper specialized
653 SLM+typed route or deterministic shared-operator execution.

654 A no-call same-operator program-trace exhibit extracts ten representative cases from the completed
655 convenience and atomic trajectory logs. The artifact covers all 45 rows for both variants, records
656 the aggregate failure-class mix and prints selected generated code blocks without quoting long input
657 contexts. It makes the same-operator mechanism inspectable beyond the separate three-row original-
658 standard-RLM trajectory subset: the convenience row contains 348 generated code blocks and the
659 atomic row contains 337, with examples of deterministic parsing successes, wrong-route choices,
660 malformed finalization, tool-use confusion and timeout/error behavior. This remains qualitative
661 mechanism evidence; the aggregate claims are still the completed 31/45 and 29/45 CD-45 rows
662 above.

663 The trace examples in Table 16 are not a substitute for aggregate statistics; they explain why the
664 aggregate statistics are plausible. They show three distinct mechanisms: standard RLM can still
665 make ordinary aggregation mistakes, direct whole-context prompting can miss structured counts
666 and SLM+typed can fail when local labels are right but the typed reduction has an output-edge case.
667 The same artifact also freezes the full CD-45 outcome-pattern counts across standard RLM/Gemini,
668 standard RLM/GPT-5, SLM+typed and direct Gemini. It avoids quoting long contexts and records
669 the trace boundary directly: the aggregate CD-45 standard-RLM runs keep row predictions, costs and
670 errors while SLM/direct paths keep prompts, raw responses and token traces.

671 To inspect generated programs directly, we ran a separate three-row standard-RLM/Gemini trajectory
672 subset on the frozen CD-45 manifest with the `rlms` logger enabled. This qualitative artifact solves

2/3 rows, costs \$0.015245, records no harness errors and captures 14 generated code blocks. The two successes illustrate different program shapes: a least-label row uses batched local LLM classification followed by a Counter while a most-frequent-user row uses regex extraction and a Counter. The failed label-comparison row also uses batched local classification but its comparison/final-answer path returns “more common than” where the gold relation is “less common than.” The subset supplies program text for mechanism inspection without changing the CD-45 aggregate accuracy, cost or paired-significance estimates.

To estimate rollout variability, we also ran a full independent standard-RLM/Gemini repeat on the same frozen CD-45 manifest, with the same standard/unlabeled context and 360-second row cap. The repeat solves 21/45 exact, has 55.8% mean score, costs \$0.970 and records two 360-second timeout/error rows. The original and repeated RLM executions are exact on the same 15 examples, wrong on the same 12 examples and disagree on 18 examples: 12 original-only exact and 6 repeat-only exact. Compared with this repeat, SLM+typed has exact discordance 17/2 and remains 36/45 exact. CD-45 therefore supports the SLM+typed advantage under a second RLM execution while the original bootstrap interval remains conditional on observed executions rather than covering both sampled contexts and stochastic scaffold rollouts. The repeat also preserves the task-conditional pattern: RLM is best on user rows (10/15 repeat exact) and weaker on counting/timeline (5/15 and 6/15), with the two errors on 262k-token timeline/user rows.

Controller and policy fidelity ladder. The following rows have different evidential roles. Standard `r1ms` with Gemini 2.5 Flash is the primary audited scaffold; GPT-5 and GPT-5.6 Luna are controller-and-stability sensitivities within that scaffold; RLM-Qwen3-30B-A3B is a bounded official trained-policy diagnostic; public SRLM-style and earlier λ -RLM adapters are pressure checks; and the unmodified official λ -RLM repository contributes a complete, rankable CD-15 cross-task row. We therefore do not pool these rows, treat model diversity as independent replication or promote the official-code row to original-paper protocol parity.

A separate CD-15 repeatability stop-rule is now complete. The subset was frozen from CD-45 before the repeat calls, balances five examples per task group and five per length bucket, uses 14 context windows with max reuse two and was selected deterministically. On the parent CD-45 outputs, that fixed subset has SLM+typed 15/15 exact at \$0.107, standard RLM/Gemini 10/15 at \$0.308, GPT-5 standard RLM 10/15 at \$0.562 with four errors and direct Gemini 9/15 at \$0.336. Three independent standard-RLM/Gemini repeats with the same standard/unlabeled context and 360-second row cap then produced 13/15 exact at \$0.513, 11/15 exact at \$0.324 and 6/15 exact at \$0.415; their combined captured OpenRouter cost is \$1.252. There are no harness timeout rows but one internal RLM error-string prediction and two free-form non-answer failures are counted incorrect. Only 3/15 examples are correct in all three repeats while one relative-frequency row is wrong in every repeat. We intentionally do not present a pooled repeat accuracy: the three outcomes per item are dependent rollout observations, not 45 independent benchmark examples. This is not a benchmark-scale variance estimate but it directly documents that the standard RLM scaffold is stochastic enough that a single small frozen subset can range from RLM-favorable to clearly weak under the same controller and row cap.

We replayed direct Gemini on the same CD-15 rows. The chronology matters: the subset was frozen for repeatability after the parent CD-45 outcomes were known and the positive-side interpretation was formulated after the RLM repeats. This is therefore a retrospective descriptive diagnostic, not a prospectively selected positive regime. The direct replay solves 8/15 exact at \$0.341 with zero errors; the three RLM rollouts solve 13/15, 11/15 and 6/15. Averaging the three RLM outcomes within each of the 15 unique items gives a 13.3-point advantage over the one observed direct replay but an item-cluster bootstrap spans −13.3 to 40.0 points. This interval is conditional on the three observed RLM rollouts and one direct rollout, not total fresh-rollout uncertainty. We do not report an aggregate McNemar test: duplicating the single direct replay three times would create 45 dependent pseudo-pairs. The stored SLM+typed route remains 15/15 exact at \$0.107.

To test whether the result depends on a generic prompted controller, we served the official MIT OASYS RLM-Qwen3-30B-A3B policy (base `Qwen/Qwen3-30B-A3B-Instruct-2507`, LoRA `mit-oasys/r1m-qwen3-30b-a3b-v0.1`) on one 80GB-class RunPod GPU and ran it through the authors’ current canonical `r1m` harness. We matched the model-card Oolong inference settings: depth one, at most 20 iterations, 4096 controller and child completion tokens, disabled Qwen thinking, the official Oolong instruction plus the natural-language question, standard/unlabeled

CD-15 route	Exact	Mean	Cost	Err.
Stored SLM+typed	15/15	100.0%	\$0.107	0
Direct Gemini replay	8/15	53.3%	\$0.341	0
Standard RLM repeat 1	13/15	86.7%	\$0.513	0
Standard RLM repeat 2	11/15	78.3%	\$0.324	0
Standard RLM repeat 3	6/15	48.8%	\$0.415	0

Table 17: Retrospective CD-15 rollout diagnostic on the frozen repeatability subset. The direct replay was observed once and the three dependent RLM rollouts are reported separately. Item-clustered uncertainty spans -13.3 to 40.0 points so this is not a powered RLM-over-direct claim. Costs are observed captured costs, not projected repeated-direct spend.

Frozen CD-15 route	Exact	Mean	Complete	Cost
Stored SLM+typed	15/15	100.0%	15/15	\$0.107 API
Stored direct Gemini	9/15	65.0%	15/15	\$0.336 API
Stored standard RLM/Gemini	10/15	70.4%	15/15	\$0.308 API
Official RLM-Qwen3-30B-A3B	10/15	70.4%	14/15	\$1.719 GPU [†]
Official-code λ -RLM/Qwen3-8B	8/15	53.3%	13/15	\$5.450 GPU [‡]

Table 18: Official-policy and official-code diagnostics on the identical frozen CD-15 manifest. [†]The trained-policy cost includes cold start and its declared 1,500-second timeout cap; finalized infrastructure billing was \$12.510 after a teardown overrun. [‡]The λ cost is the creation-to-cleanup listed-rate estimate including two 1,800-second caps; finalized infrastructure billing was \$5.471. Neither row is a full SRLM/ λ -RLM/RAH benchmark reproduction.

729 context_window_text and the same local scorer used by the stored CD-15 baselines. The local
730 official-harness checkout was fixed at commit 72d6940142ddfb84ee6be573dc999a37e633e671.
731 No task metadata, local gold labels or typed Oolong operators were supplied.

732 The policy returns 10/15 exact and 70.4% mean Oolong score. It completes 14/15 rows; the final
733 262k-window relative-frequency row returns no response within the predeclared 1,500-second outer
734 cap and is scored incorrect. By task group it reaches 3/5 exact on counting, 4/5 on timeline and
735 3/5 on user. The 14 completed trajectories contain 70 iterations and 70 generated code blocks;
736 completed-row median latency is 15.3 seconds and maximum latency is 72.9 seconds. Against
737 SLM+typed, paired exact discordance is 0 official-only versus 5 SLM-only ($p = 0.0625$); against
738 stored direct Gemini it is 5 versus 4 ($p = 1.0$) and against stored standard RLM/Gemini it is 4 versus
739 4 ($p = 1.0$). The official policy, stored standard RLM/Gemini and stored standard RLM/GPT-5 each
740 reach 10/15 exact on this subset. The diagnostic adds official trained-policy coverage and shows
741 a numerical RLM-over-direct difference, but it neither establishes a powered win nor reverses the
742 specialized route on this slice.

743 The deployment log separates scientific protocol cost from an operational incident. At the listed
744 \$2.99/hour rate, elapsed GPU time from pod rental through cold start, 14 completions and the final
745 timeout cap is \$1.719. Finalized RunPod billing was \$12.510 because the final request outlived the
746 local alarm and the detached pod remained active after the local process died; the \$10.791 difference
747 is teardown overrun, not model compute required by the declared protocol. OpenRouter spend for this
748 run was \$0.00. The anonymous release contains the adjudicated row table, analysis JSON/Markdown
749 and compact generated-program trajectories; the 105MB raw trajectory file is retained locally rather
750 than included in the compact release bundle.

751 The completed GPT-5 controller run does not eliminate the CD-45 difference. SLM+typed beats
752 GPT-5 standard RLM on paired exact outcomes ($b = 15, c = 2$, two-sided exact $p = 0.00235$), with
753 a context-cluster exact-difference interval of 11.1–47.7 points. GPT-5 is not significantly different
754 from the logged Gemini 2.5 Flash standard-RLM row on paired exact outcomes (GPT-5-only 6,
755 Gemini-only 10, $p = 0.454$) but it is less stable under the strict stop rule: 15/45 rows record timeout
756 or scaffold-error failures. Because the run changes the top-level controller while retaining the local
757 scaffold and not recreating the original GPT-5/GPT-5-mini nested-call configuration, it should be read
758 as evidence that a stronger top-level controller alone does not remove the standard scaffold’s bounded-

CD-45 route	Exact	Mean	Cost	Err.
Q-parsed SLM+typed	36/45	83.7%	\$0.416	0
Standard RLM / Gemini 2.5 Flash	27/45	62.3%	\$1.001	0
Standard RLM repeat / Gemini 2.5 Flash	21/45	55.8%	\$0.970	2
Standard RLM + same typed ops / Gemini 2.5 Flash	31/45	69.3%	\$1.314	2
Standard RLM / GPT-5	23/45	58.1%	\$2.442	15
Standard RLM / GPT-5.6 Luna	26/45	62.9%	\$3.966	2

Table 19: Completed strict-stop CD-45 controller-freshness sensitivity. The GPT-5 row uses openai/gpt-5 via OpenRouter inside the same local standard-rlms scaffold with a 240-second per-row timeout; the GPT-5.6 Luna row uses the central 360-second cap. These are controller/stability sensitivities, not replacements for official SRLM or λ -RLM reproductions.

Variant	Robust	Strict	Exact	Mean
Original	120/120	120/120	80/120	72.7%
Answer format removed	120/120	29/120	80/120	72.7%
Deterministic paraphrase	120/120	0/120	80/120	72.7%

Table 20: No-call Oolong question-paraphrase stress test over the same frozen $N = 120$ rows. The robust parser receives only the question variant and stored SLM row-label predictions; it does not use benchmark task/output metadata. “Strict parser ok” reports whether the original format-template parser can infer task/output schema from the variant.

Len.	SLM	RLM	SLM \$	RLM \$	Ratio
8k	16/24	14/24	\$0.0199	\$0.2528	12.7×
16k	19/24	12/24	\$0.0414	\$0.2356	5.7×
32k	16/24	13/24	\$0.0852	\$0.4125	4.8×
65k	15/24	4/24	\$0.1768	\$0.4464	2.5×
131k	14/24	10/24	\$0.3357	\$1.3410	4.0×

Table 21: Length-sliced exact rates and captured successful-completion API costs for the $N = 120$ standard-unlabeled replication. Each length bucket has 24 examples. Recorded standard-RLM scaffold errors by bucket are 0, 0, 1, 5 and 3 respectively.

run reliability problem on this slice, not as evidence against all frontier-controller RLM-family variants.

Current-controller sensitivity. A complete GPT-5.6 Luna standard-RLM row uses the same 360-second cap as the central Gemini row. Luna completes all 45 rows: 26/45 exact, 62.9% mean score, two timeout/error rows and \$3.965806 captured successful-completion cost. SLM+typed remains higher on the same rows (36/45 exact, \$0.416415), with paired discordance $b = 15$, $c = 5$, two-sided exact $p = 0.041$ and a 32-cluster exact-difference interval of 2.3–40.9 points. Luna is statistically indistinguishable from the logged Gemini 2.5 standard-RLM row (Luna-only 10, Gemini-only 11, $p = 1.0$), directionally above GPT-5 (12 versus 9 discordant exact wins, $p = 0.664$) and directionally above direct Gemini (12 versus 7, $p = 0.359$). It does not reverse the operation-matched SLM+typed decision boundary and is about $9.5\times$ costlier than SLM+typed on captured successful-completion cost.

Table 20 is not a new model run; it is a parser/harness stress test. It shows that the original exact-format parser is template-sensitive while a slightly more semantic rule parser still recovers the task and answer-type fields from format-removed and paraphrased questions. This reduces but does not eliminate the Oolong-template concern: the paraphrases are deterministic perturbations of Oolong questions, not independently authored questions from a new benchmark.

Each length bucket has only 24 examples, so we do not interpret the slice as a leaderboard. SLM-labeler+typed remains ahead in every bucket while both accuracy and cost change with context

Noise view	Count	Interpretation
SLM local-label/model noise	20	SLM+typed wrong with row-label accuracy < 0.90 ; cheap workers are strong but not solved readers.
SLM aggregation sensitivity	20	SLM+typed wrong despite row-label accuracy ≥ 0.90 ; small local errors or schema edge cases can flip exact aggregates.
RLM scaffold/reliability noise	9	Captured RLM timeout/scaffold errors, concentrated at 65k/131k windows.
Direct whole-context model noise	66	Direct Gemini wrong without infrastructure errors; removing recursion alone does not solve aggregation.
RLM non-error answer noise	58	RLM wrong without captured scaffold error; most misses are ordinary wrong answers plus a reliability tail.

Table 22: No-call noise-decomposition diagnostic for the $N = 120$ standard-unlabeled Oolong replication, following the divide-and-conquer view that failures can arise from task/local interpretation, length/model degradation, aggregation or scaffold reliability. Categories are operational diagnostics over frozen outputs, not new gold labels.

length; the captured RLM/SLM cost multiplier therefore persists into the longer evaluated buckets. At 131k, SLM-labeler+typed still beats standard RLM but falls to 14/24 exact, showing that cheap local labeling does not solve the benchmark. RLM is weakest at 65k (4/24 exact, five errors) and most expensive at 131k (\$1.341 for 24 examples). Attempted-cost auditing remains conservative: the stored RLM logs contain 85 official-RLM attempt rows, 76 completions and 9 error rows, all with zero recorded failed-call usage so reported attempted cost is a lower bound rather than an imputed provider bill for failed partial executions. The two-method and three-way score/cost PDFs are included in the released figures/ directory; the three-way files use the n120_threeway_score_by_length and n120_threeway_cost_by_length names.

The cleaner CD-45 split-validity slice gives the complementary long-context cost check at 1k/4k/262k windows. On those rows, SLM+typed costs \$0.00095, \$0.00610 and \$0.40937 by length and solves 13/15, 11/15 and 12/15. Standard RLM costs \$0.09659, \$0.12127 and \$0.78316 and solves 12/15, 8/15 and 7/15. Direct Gemini is cheap at 1k/4k (\$0.00352/\$0.01509) but becomes the most expensive path at 262k (\$1.00599) while solving 4/15. Giving standard r1ms the same typed operator library helps but does not remove the cost-scale pattern: the convenience row costs \$0.13746/\$0.17250/\$1.00449 at 1k/4k/262k and the atomic-only row costs \$0.17118/\$0.20065/\$0.58686, with long-row timeout/error counts of 2 and 4. Together, the two public-contract Oolong ladders cover 8k–131k and a context-window-ID-disjoint 262k stress point under explicitly different overlap properties.

The failure modes separate in a way that matches recent long-context decomposition work. RLM still beats direct/CoT Gemma overall in the $N = 30$ four-method pilot and wins several user-ID cases where whole-context Gemma copies a source row or chooses the wrong entity, but the $N = 120$ run adds nine timeout/scaffold errors and many wrong numeric or wrong-output-type predictions. Direct Gemini avoids those scaffold errors and is cheaper but still has 66 non-error wrong answers. SLM-labeler+typed has no scaffold errors, yet its 40 misses split evenly between local-label/model noise and aggregation sensitivity after high-accuracy local labeling. The evidence does not show that SLM solves Oolong. Instead, it shows that the dominant failure allocation changes by mechanism and that the current standard-RLM scaffold does not reduce model or aggregation noise enough to justify its extra cost on this frozen public contract. The diagnostic JSON, CSV and PDF artifacts are included in the release under the Oolong N=120 noise-decomposition filenames.

Route	Complete	Correct G/S	Median / max s	Active \$
Standard RLM	42/45	33/45 / 33/45	44.6 / 1,289.4	6.746
Text decomposition + same Qwen	45/45	29/45 / 28/45	29.6 / 62.5	1.610
BM25 + same Qwen	42/45	21/45 / 21/45	3.2 / 21.2	0.294

Table 23: Disjoint fixed- $N=45$ BrowseComp-Plus replication. G/S reports Gemma/Sol correct counts; Gemma Wilson intervals are 59.0–84.0, 49.8–76.8 and 32.9–60.9 for RLM, text and BM25. Failures count incorrect. RLM–BM25 is +26.7 points ($b=15, c=3$, paired bootstrap 11.1–44.4, exact $p=0.0075$, two-contrast Holm $p=0.0151$). RLM–text is +8.9 points ($b=9, c=5$, -6.7 – 24.4 , $p=0.424$) and remains uncertain.

F BrowseComp-Plus Fixed- N Replication

To remove the stopping-rule ambiguity from the positive-regime endpoint, we froze $N=45$ rows from official query shard 1 before output. The manifest uses SHA-256 ordering, contains exactly 1,000 unique documents per row and has zero query-ID or query-hash overlap with the earlier $N=80$ and $N=59$ manifests. Model revision, official r1ms commit, three route prompts, construction, caps, failure-as-incorrect policy and primary judge are unchanged. A local-driver interruption occurred after 107/135 cells. Before score inspection, a numbered infrastructure amendment preserved those cells byte-for-byte and resumed exactly the 28 missing cells on the surviving pinned pod. The combined compactor verifies 107 plus 28 non-overlapping cells, all 135 attempted cells and both raw hashes; the pod was deleted before scoring.

The primary Gemma 4 judge attempted all 129 successful generations; 128 responses parsed and one format failure was conservatively scored incorrect. GPT-5.6 Sol at high reasoning then re-judged all 129 outputs under route and primary-label blinding. Agreement is 128/129 (99.2%, Cohen’s $\kappa=0.983$). Sol preserves RLM and BM25 totals, changes text from 29 to 28 and leaves the two RLM conclusions unchanged. This is independent second-model sensitivity, not human validation. Generation and primary judging cost \$10.120 and \$0.338 at listed RunPod rates. Sol returned \$0.590893 in per-call cost while the concurrent account delta was \$0.496861; both are retained because the credential may be shared. Restricted model I/O stays local. The release contains the public manifest, protocol, amendment, release-safe aggregates, source hashes and an independent 135-cell verifier.

G BrowseComp-Plus Equal-Cap Comparison and Rollout Sensitivity

The strongest BrowseComp-Plus endpoint prospectively freezes 45 further untouched shard-1 queries before any route output and attempts all 180 cells. Standard RLM, iterative lexical search, textual decomposition and BM25 use the same self-hosted Qwen3.6-35B-A3B-FP8 checkpoint and seed 20. Each route has at most 20 sequential decisions and 1,200 seconds per row; route-specific access patterns differ by design. The execution required one qualification row, an interrupted immutable ranks 2–19 episode and a hash-bound pre-score ranks 20–45 recovery. Failures remain incorrect in the fixed denominator.

Route	Comp. / corr.	Wilson 95%	p50/p95 Calls	seconds	\$
RLM	41/31	54.3–80.5	476	50.1/628.8	9.873
Iterative	45/33	59.0–84.0	426	14.1/90.2	1.710
Text decomp.	45/37	68.7–90.7	311	22.1/47.6	1.400
BM25	42/25	41.2–69.1	42	3.0/20.0	0.296

Table 24: Prospective equal-cap BrowseComp-Plus endpoint, seed 20 ($N = 45$). Entries under “Comp./corr.” are completed/correct counts. Costs are route-attributed active runtime; the four routes sum to \$13.279. Adding the interrupted-episode billing lower bound and primary judging gives \$15.015.

RLM contrast	Diff. Only	95% interval / exact p	Holm
Iterative search	−4.4 6 / 8	[−20.0, +11.1] / .7905	.7905
Text decomposition	−13.3 5 / 11	[−31.1, +4.4] / .2101	.5387
BM25 + Qwen	+13.3 10 / 4	[−2.2, +28.9] / .1796	.5387

Table 25: Declared seed-20 paired RLM contrasts in percentage points. “Only” gives RLM-only/other-only correct counts. Holm adjustment covers the three comparisons; none rejects and the minimum adjusted value is 0.5387.

On the 39 rows completed by every route, RLM/iterative/text/BM25 score 30/28/31/22. This common-completion analysis is post-treatment and does not replace the all-attempted endpoint. GPT-5.6 Sol independently re-judges all 173 successful outputs under route and primary-label blinding and agrees on 172/173 ($\kappa=0.985$). This is second-model scorer stability, not human validation.

Post-hoc seed-21 repeat. The same 45 rows were rerun with only the model-card seed changed from 20 to 21. This same-row sensitivity was specified after the seed-20 result, is not an independent endpoint and is not pooled with it.

Route	Raw/cap / corr.	Wilson 95%	p50/p95 seconds	\$
RLM	37/36/30	52.1–78.6	64.2/1,219.4	18.170
Iterative	45/45/31	54.3–80.5	7.9/68.3	1.145
Text decomp.	45/45/29	49.8–76.8	26.0/44.8	1.498
BM25	42/42/22	35.0–63.0	3.2/21.0	0.356

Table 26: Post-hoc seed-21 same-row rollout sensitivity ($N = 45$). “Raw/cap/corr.” gives raw completion, cap-valid completion and correct counts. Seven RLM rows exceed the 1,200-second cap; the sole late nonempty output is incorrect, so fail-closed scoring leaves 30/45 unchanged.

RLM versus iterative/text/BM25 differs by −2.2, +2.2 and +17.8 points, with Holm-adjusted p values 1.0, 1.0 and 0.346. Route ordering changes but the non-significant RLM contrast set repeats. On the 34 rows cap-valid for every route, RLM/iterative/text/BM25 score 28/23/20/18; this is again a post-treatment sensitivity. RLM active-runtime cost is $15.9\times$ iterative search and $12.1\times$ textual decomposition. The four route-attributed totals sum to \$21.170 and generation plus primary judging costs \$22.374 from pod creation through cleanup. Sol agrees on 166/169 successful outputs ($\kappa=0.961$). The seed repeat supports rollout-instability caution, not equivalence or independent replication.

Dominance, transitions and completion across rollouts. After freezing the equal-cap contract, we recomputed row-level correctness from the retained compact generations and primary Gemma 4 judgments without new model calls. Published seed-20 counts 31/33/37/25 and seed-21 counts 30/31/29/22 reconstruct exactly. Under accuracy-up/cost-down dominance, seed 20 retains only textual decomposition and BM25 as non-dominated; RLM and iterative search are dominated. Seed 21 flips the adaptive frontier: iterative search and BM25 remain non-dominated while textual decomposition becomes dominated, matching the main-text observation that route ordering is rollout-sensitive. Mean per-row accuracy regret against an oracle that picks any correct route on that row is lowest for text on seed 20 (0.156) but rises to 0.289 on seed 21, whereas RLM regret stays near 0.27–0.29 and BM25 remains highest (≈ 0.42 –0.44).

These panels are descriptive re-analyses of the frozen equal-cap evidence. They do not replace the Holm-controlled paired tests, do not add a prospective dense full-pool route and do not authorize pooling seed 20 with seed 21.

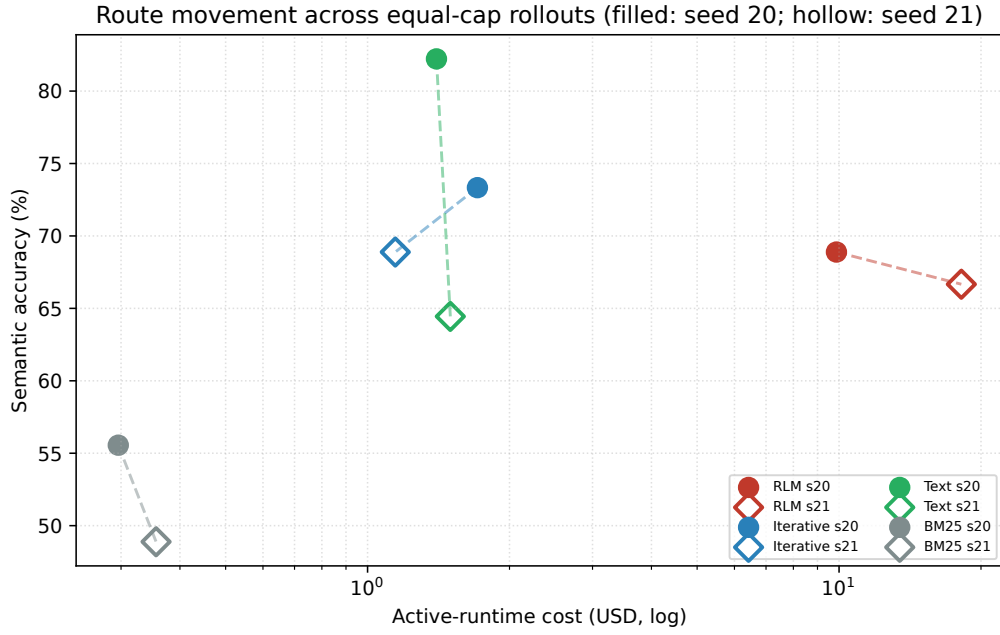


Figure 3: Equal-cap accuracy–cost movement across rollouts (filled markers: prospective seed 20; hollow markers: post-hoc seed 21). Lines connect the same route. Costs are route-attributed active runtime on a log axis. The plot does not pool seeds into one estimator.

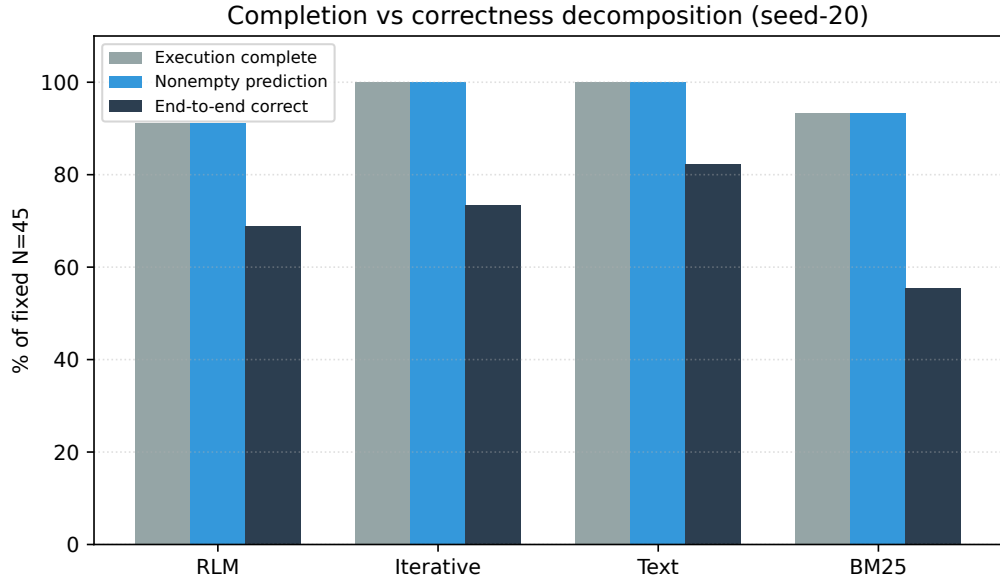


Figure 4: Completion-versus-correctness decomposition on the prospective seed-20 endpoint: execution completion, nonempty prediction and end-to-end semantic correctness under the fixed $N=45$ denominator. RLM’s gap is concentrated in completion and end-to-end correctness rather than empty predictions alone.

Route	s20	s21	TT	FT	TF	FF
RLM	31	30	23	7	8	7
Iterative	33	31	27	4	6	8
Text decomp.	37	29	28	1	9	7
BM25	25	22	22	0	3	20

Table 27: Seed-20→seed-21 correctness transitions on the same 45 rows (TT=stable correct, FT=gained, TF=lost, FF=stable incorrect). Text loses nine previously correct rows and gains one; RLM is comparatively stable in total correct count but still cost-dominated. This is rollout stability, not an independent task sample.

H Repeatability and Cost Accounting

Controlled families. T-STRUCT and the definitive T-MIXED $N = 30$ comparison are single full-family passes. The T-MIXED old-finalizer run is a same-model $N = 20$ mechanism probe; its exact-example manifest is reconstructed from stored model-I/O traces. The current-contract extension instead uses a separately frozen stable-generator manifest and SHA-256-derived offset, avoiding the historical generator’s salted Python hash(domain) seed.

Public anchors. BrowseComp-Plus contributes four major estimates: the frozen $N = 80$ two-episode completion, the zero-overlap score-blind attempted- $N = 49$ replication stopped by its stability rule, the disjoint fixed- $N = 45$ three-route replication with all 135 cells attempted before scoring and the prospective fixed- $N = 45$ four-route equal-cap endpoint with all 180 cells attempted before scoring. The seed-21 rerun is a post-hoc same-row rollout sensitivity, not a fifth independent estimate. The earlier $N = 37$ prefix records the first study’s interruption and is not a confirmatory sample. T-NLP and T-SQ remain compatibility/proxy diagnostics; LB2 >500k and HotpotQA are bounded public diagnostics; and the public λ -RLM andprehend SRLM-style rows are pressure tests rather than official reproductions.

Cost accounting. Cost ratios compare routes within task families rather than through a global method denominator. RLM totals cover logged successful completions. The attempted-cost audit records their lower bound and the zero-usage failed rows that prevent exact reconstruction of partial failed-call billing.

Models and execution. Hosted models are identified by dated provider slug in the experiment records. The self-hosted BrowseComp endpoint uses Qwen3.6-35B-A3B-FP8 at a frozen weight revision and serving-image digest, a 65,536-token context limit, model-card decoding, disabled thinking and seed 20 (or 21 only for the labeled repeat). The official λ row pins repository commit 3874d393483d, weight revision b968826d9c46 and vllm/vllm-openai:v0.21.0 on one 80GB H100. The official trained-policy row uses an 80GB-class RunPod GPU and the canonical rlm harness at commit 72d6940142ddf84e6be573dc999a37e633e671.

Compute accounting. API experiments log returned tokens and list-price cost. Self-hosted BrowseComp costs use route-attributed active runtime at the recorded pod rate; generation-plus-judge totals are reported separately. Failed hosted calls sometimes return no usage, so attempted API cost is a lower bound. The complete official λ CD-15 row required 6,562 seconds on one 80GB H100 and cost \$5.471 from creation through cleanup. The prospective seed-20 equal-cap endpoint attributes \$13.279 across route runtime and \$15.015 including the interrupted-episode lower bound and primary judging. The seed-21 repeat attributes \$21.170 across routes and \$22.374 from pod creation through primary judging and cleanup.

Data and release boundary. The study uses public or synthetic benchmark interfaces and does not redistribute source-document-scale BrowseComp-Plus, Oolong, LongBench-v2 or HotpotQA corpora. Access remains governed by upstream terms. The project retains frozen manifests, prompts, aggregate outputs, model-I/O traces, cost ledgers and deterministic claim verifiers. A public artifact is planned after anonymous review; the current single-PDF workshop submission makes no claim that code or data are already openly downloadable.

I Reproducibility Commands

Integrity and build commands. From the extracted release root, run `python3 verify_release.py`; success is the final ANONYMOUS RELEASE VERIFIER: PASS line. This deterministic command requires no network, model, API or GPU access.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

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3. Theory assumptions and proofs

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Justification: Section 2 and Appendix H specify inputs, routes, scoring, failures, uncertainty, versions, caps, compute and cost boundaries for the claim-bearing results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No].

Justification: The single-PDF workshop submission does not provide a public code/data URL. Appendix H documents the reproduction contract and asset boundaries; an anonymized artifact is planned after review.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: Section 2 specifies the frozen splits, route roles, decoding policies, seeds, estimands, scorers and stopping rules. No model is trained by this work; official trained checkpoints are inference-only diagnostics.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes].

Justification: Section 2 defines paired McNemar tests, fixed-seed row and cluster bootstraps and Holm correction; Section 3 and Appendix G report intervals, discordances and adjusted values for central contrasts.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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Justification: Appendix H identifies hosted APIs, self-hosted model revisions, 80GB GPU deployments, container/harness versions, elapsed time and route-level costs for the principal self-hosted runs.

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998 Question: For crowdsourcing experiments and research with human subjects, does the paper
 999 include the full text of instructions given to participants and screenshots, if applicable, as
 1000 well as details about compensation (if any)?

1001 Answer: [N/A].

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 1007 approvals (or an equivalent approval/review based on the requirements of your country or
 1008 institution) were obtained?

1009 Answer: [N/A].

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